

1. Implement the following function using

i) 2:1 Mux and NOT gate

ii) 4:1 MUX

a) $F(A, B, C) = \bar{A}C + B\bar{C}$ (∵ can't use NOT gate)

A B C F

0 0 0 0

0 0 1 1

0 1 0 1

0 1 1 1

1 0 0 0

1 0 1 0

1 1 0 1

1 1 1 0

C as input

A B F

0 0 C

0 1 1

1 0 0

1 1 $\bar{C}$

B as input

A C F

0 0 B

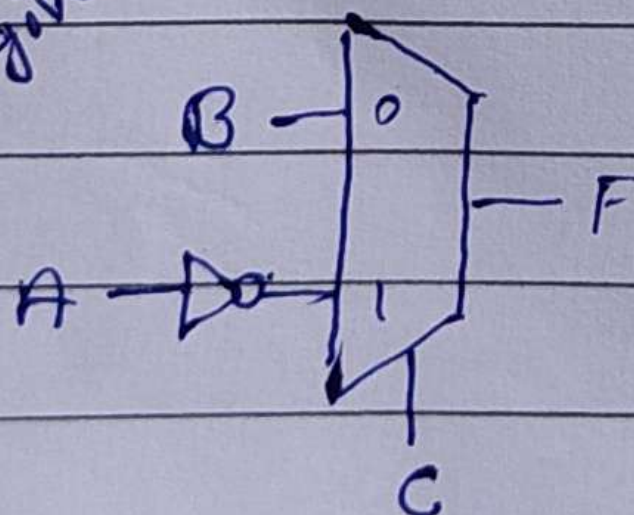
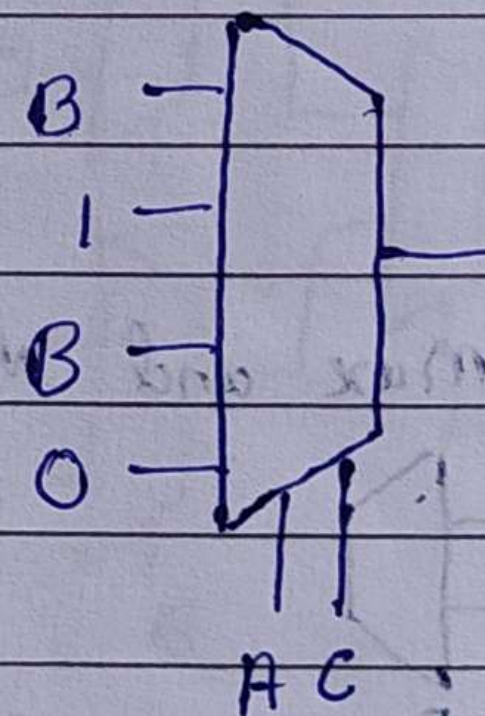
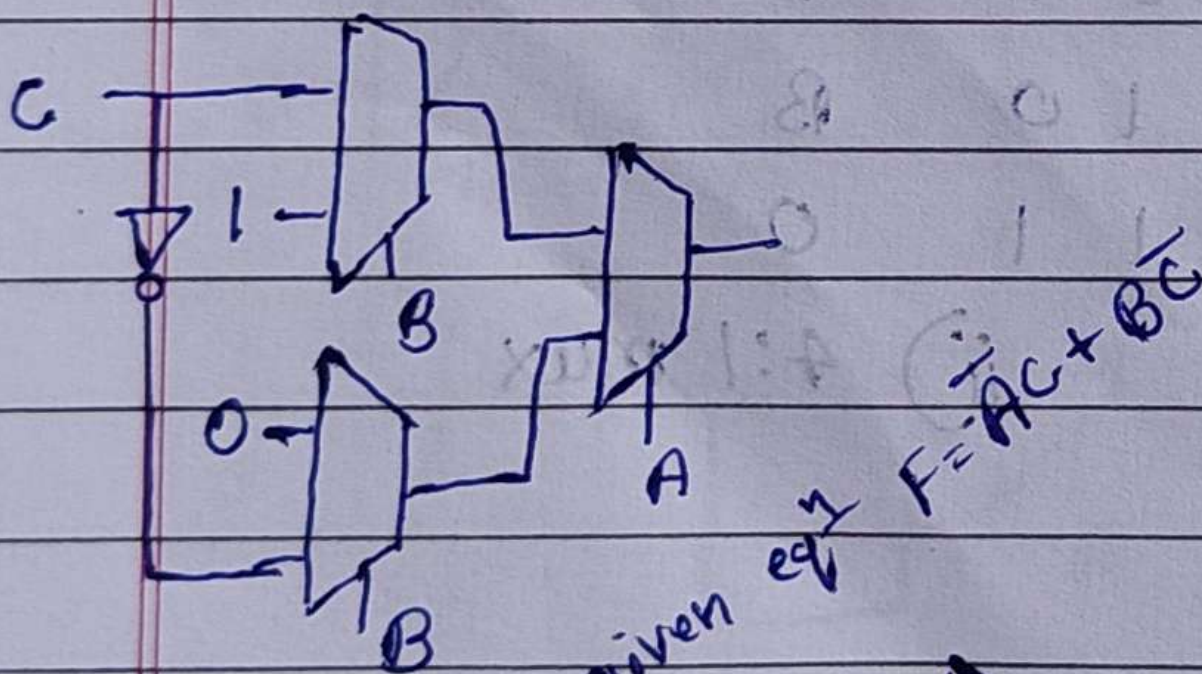
0 1 1

1 0 B

1 1 0

i) 2:1 MUX and NOT gate

ii) 4:1 MUX



$$b) F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$$

P	Q	R	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

R as input

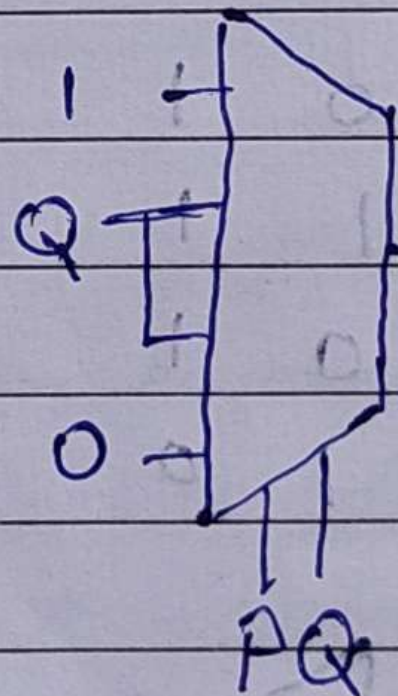
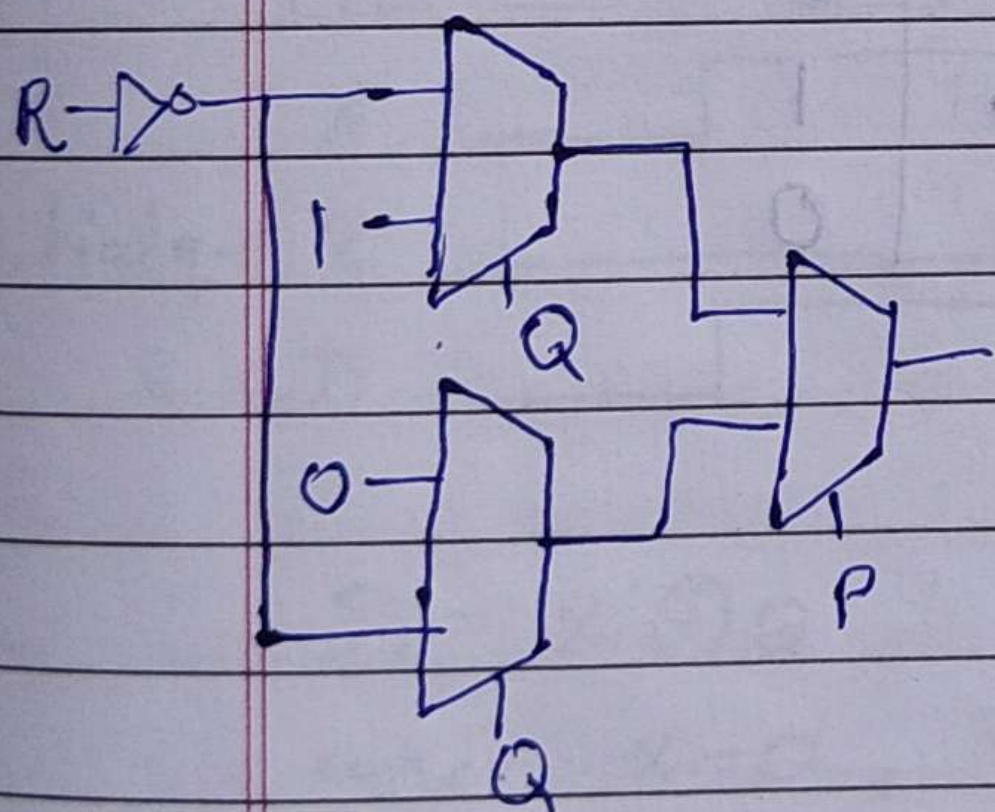
P	Q	F
0	0	$\bar{R}$
0	1	X
1	0	0
1	1	$\bar{R}$

Q as input

P	Q	F
0	0	1
0	1	Q
1	0	Q
1	1	0

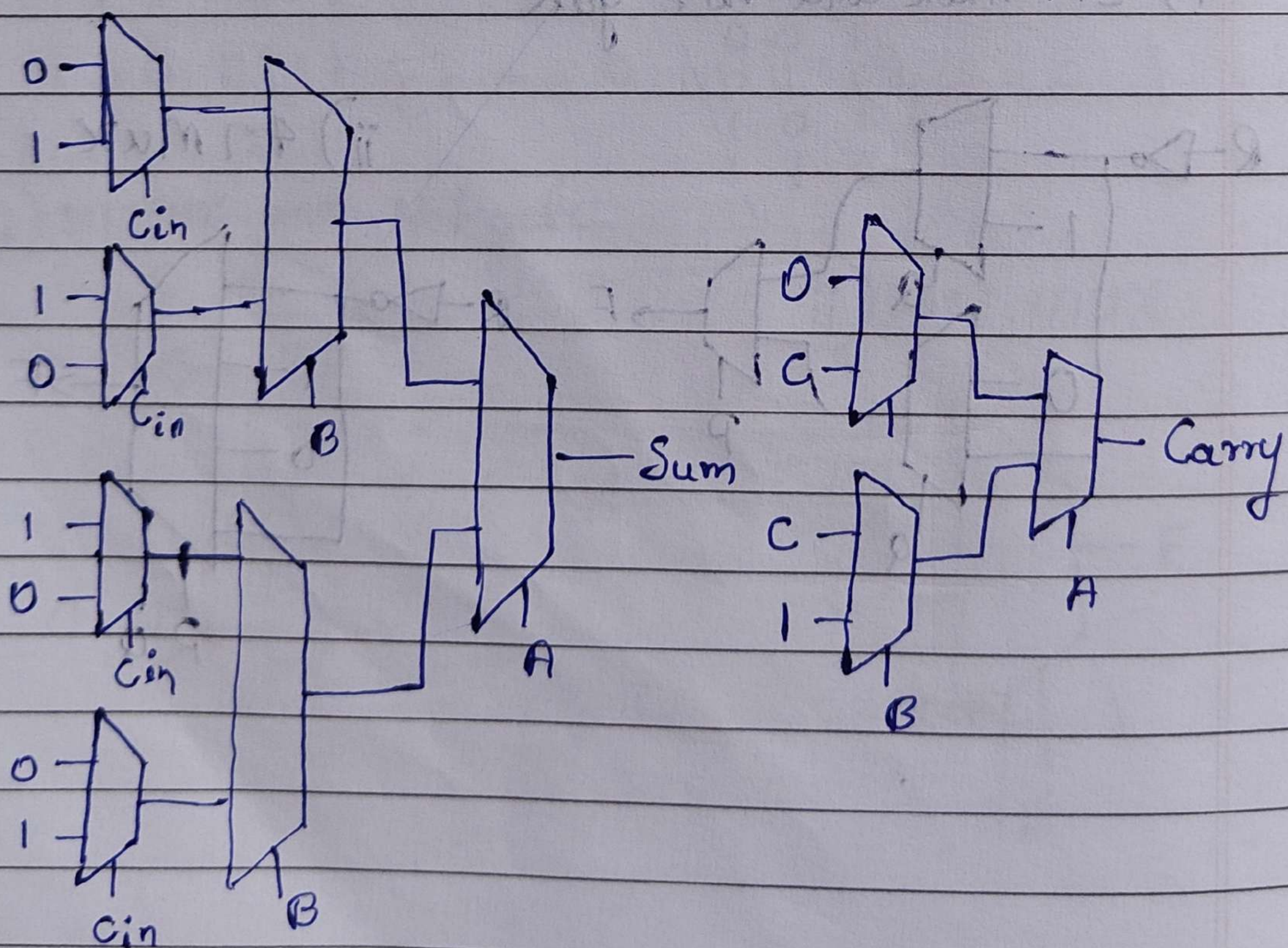
i) 2:1 MUX and NOT gate

ii) 4:1 MUX



2. Implement the functionality of a full adder using 2:1 Muxes only.

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



3. Simplify $f(a,b,c,d) = \sum m(2,4,5,6,10) + D(12,13,14,15)$ and implement the SOP expression using 4:1 MUX.

f	cd				
ab	00	01	11	10	
00				1	
01	1	1		1	
11	X	X	X	X	
10				1	

$$f = b\bar{c} + c\bar{d} \quad (\text{SOP})$$

b c d f

0 0 0 0

0 0 1 0

0 1 0 1

0 1 1 0

1 0 0 1

1 0 1 1

1 1 0 1

1 1 1 0

$f = b\bar{c} + c\bar{d}$ $\therefore$ b as input for Mux
to avoid NOT gate

c d f

0 0 b

0 1 b

1 0 1

1 1 0

